Introduction to Machine Learning
Homework 5

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Question 1

Generally, solving a classification problem using a linear regression model on a one-hot indicator matrix $\boldsymbol{Y}$ does not work well. However, it can perform optimally under specific strict conditions.

When it does not work well (The Masking Problem):
According to *The Elements of Statistical Learning* (ESL), when the number of classes $C \geq 3$, the linear regression model often suffers from the **masking effect**. Because the model forces a rigid linear fit to the one-hot vectors, the predicted values are not true probabilities. If the classes are approximately linearly aligned in the feature space, an "intermediate" class may be completely masked by its surrounding classes. Its predicted value will never be the maximum among all classes, meaning the $\arg\max$ decision rule will never assign a sample to this class.

When it works well (Equivalence to LDA):
Least squares classification theoretically and empirically performs well in the following scenarios:

- **Binary Classification ($C = 2$):** When there are only two classes, there is no intermediate class, ensuring the masking effect cannot occur.
- **Equal Covariance and Balanced Priors:** From a probabilistic generative perspective, least squares fitting on one-hot labels performs optimally when the underlying data distribution satisfies the assumptions of Linear Discriminant Analysis (LDA):
  1. **Homoscedastic Gaussians:** Each class-conditional density is a Gaussian sharing the identical covariance matrix Σ: $p(x \mid y = k) = \mathcal{N}(x; \mu_k, \Sigma)$, $\forall k$.
  2. **Equal Class Priors:** The classes are balanced, meaning $\Pr(y = k) = 1/C$.

  Under these assumptions, the Bayes optimal decision boundaries are affine hyperplanes. In this scenario, least squares on one-hot labels perfectly recovers the exact same decision boundary as LDA, yielding the true optimal classifier.

Question 2

Yes, we can solve classification problems using a nonlinear regression model. A prime example is formulating **Logistic Regression** strictly within the prescribed nonlinear regression framework.

1. Nonlinear Model Formulation:
Let the binary classification labels be $y \in \{0, 1\}$. According to the nonlinear regression format

$y = f_{\boldsymbol{\theta}}(\boldsymbol{x}) + \epsilon$, we define our nonlinear parametric function by passing a linear combination of features through the sigmoid nonlinear activation function:

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \frac{1}{1 + \exp(-(\boldsymbol{w}^T \boldsymbol{x} + b))} \tag{1}$$

where $\boldsymbol{\theta} = \{\boldsymbol{w}, b\}$ represents the model parameters. The function $f_{\boldsymbol{\theta}}(\boldsymbol{x})$ smoothly outputs values in $(0, 1)$, acting as the conditional expectation $\mathbb{E}[y \mid \boldsymbol{x}] = P(y = 1 \mid \boldsymbol{x})$.

2. The Meaning of the Error Term ϵ:

In classical regression, the error term ϵ is often assumed to be continuous Gaussian noise. However, in this classification setting where y is a Bernoulli random variable, the noise term $\epsilon = y - f_{\boldsymbol{\theta}}(\boldsymbol{x})$ captures the precise residual between discrete class labels and the continuous nonlinear regression curve. It is a zero-mean ($\mathbb{E}[\epsilon \mid \boldsymbol{x}] = 0$) discrete random variable:

$$\epsilon = \begin{cases} 1 - f_{\boldsymbol{\theta}}(\boldsymbol{x}), & \text{with probability } f_{\boldsymbol{\theta}}(\boldsymbol{x}) \quad (\text{when } y = 1) \\ -f_{\boldsymbol{\theta}}(\boldsymbol{x}), & \text{with probability } 1 - f_{\boldsymbol{\theta}}(\boldsymbol{x}) \quad (\text{when } y = 0) \end{cases} \tag{2}$$

3. Learning Task (Optimization Objective):

Aligned directly with the nonlinear regression paradigm, we learn the optimal parameters $\boldsymbol{\theta}$ by minimizing the empirical risk under the L_2 loss (Least Squares), exactly matching the specified learning task:

$$\min_{\boldsymbol{\theta}} \sum_{n=1}^{N} |y_n - f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)|^2 = \min_{\boldsymbol{\theta}} \|\boldsymbol{y} - f_{\boldsymbol{\theta}}(\boldsymbol{X})\|_2^2 \tag{3}$$

4. Performing Classification via Thresholding:

Once the optimal parameters $\boldsymbol{\theta}$ are learned through the nonlinear regression task, we can solve the classification problem by thresholding the continuous predicted values:

$$\text{Predicted Class} = \begin{cases} 1, & \text{if } f_{\boldsymbol{\theta}}(\boldsymbol{x}) \geq 0.5 \\ 0, & \text{if } f_{\boldsymbol{\theta}}(\boldsymbol{x}) < 0.5 \end{cases} \tag{4}$$

Question 3

The soft-margin SVM primal with parameter $\lambda > 0$:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|_2^2 + \frac{1}{2N\lambda} \sum_{n=1}^{N} \xi_n \quad \text{s.t.} \quad 1 - \xi_n - y_n(w^T x_n - b) \leq 0, \quad -\xi_n \leq 0.$$

Introduce Lagrange multipliers $c_n \geq 0$, $\tau_n \geq 0$:

$$L(w, b, \xi, c, \tau) = \frac{1}{2}\|w\|_2^2 + \sum_{n=1}^{N} \Big(\frac{1}{2N\lambda} \xi_n + c_n (1 - \xi_n - y_n(w^T x_n - b)) - \tau_n \xi_n \Big). \tag{5}$$

The dual: $\max_{c,\tau \geq 0} \min_{w,b,\xi} L$.

Inner minimization — set gradients to zero:

$$\nabla_w L = w - \sum_n c_n y_n x_n = 0 \implies w = \sum_n c_n y_n x_n, \tag{6}$$

$$\nabla_b L = \sum_n c_n y_n = 0 \implies \sum_n c_n y_n = 0, \tag{7}$$

$$\nabla_{\xi_n} L = \tfrac{1}{2N\lambda} - c_n - \tau_n = 0 \implies c_n + \tau_n = \tfrac{1}{2N\lambda} \implies 0 \leq c_n \leq \tfrac{1}{2N\lambda}. \tag{8}$$

Substitute (6)–(8) back into (5):

$$\tfrac{1}{2}\|w\|^2 = \tfrac{1}{2}\Big\|\sum_n c_n y_n x_n\Big\|^2 = \tfrac{1}{2}\sum_{n,m} c_n c_m y_n y_m (x_n^T x_m), \tag{9}$$

$$\sum_n c_n y_n (w^T x_n) = \Big(\sum_n c_n y_n x_n^T\Big) w = w^T w = \|w\|^2, \tag{10}$$

$$\sum_n c_n y_n b = b \sum_n c_n y_n \overset{(7)}{=} 0, \tag{11}$$

$$\sum_n \xi_n \big(\tfrac{1}{2N\lambda} - c_n - \tau_n\big) \overset{(8)}{=} 0. \tag{12}$$

Collecting terms from (5) and applying (9)–(12):

$$\begin{aligned}
\min_{w,b,\xi} L &= \tfrac{1}{2}\|w\|^2 + \sum_n c_n - \sum_n c_n y_n (w^T x_n) + b \sum_n c_n y_n + \sum_n \xi_n \big(\tfrac{1}{2N\lambda} - c_n - \tau_n\big) \\
&= \tfrac{1}{2}\|w\|^2 + \sum_n c_n - \|w\|^2 + 0 + 0 = \sum_{n=1}^N c_n - \tfrac{1}{2}\|w\|^2 \\
&= \sum_{n=1}^N c_n - \tfrac{1}{2}\sum_{n=1}^N \sum_{m=1}^N y_n c_n (x_n^T x_m) y_m c_m. \qquad (13)
\end{aligned}$$

Final dual problem (eliminate τ_n via $c_n \le \frac{1}{2N\lambda}$):

$$\boxed{\begin{aligned}
\max_{\{c_n\}_{n=1}^N} \quad & \sum_{n=1}^N c_n - \frac{1}{2}\sum_{n,m} y_n c_n (x_n^T x_m) y_m c_m \\
\text{s.t.} \quad & \sum_{n=1}^N c_n y_n = 0, \quad 0 \le c_n \le \frac{1}{2N\lambda}.
\end{aligned}}$$